Software Requirements Specification (SRS)

E-Commerce System

1. Introduction

1.1 Purpose

This document provides a detailed description of the requirements for the E-Commerce System. It includes functional and non-functional requirements, system features, and user interactions.

1.2 Scope

The E-Commerce System allows users to browse products, add them to a shopping cart, place orders, and make payments online. Administrators can manage products, users, and orders.

The system will be developed using:

- React (Frontend)
- Spring Boot (Backend)
- MySQL (Database)

1.3 Definitions

- **API**: Application Programming Interface
- **JWT**: JSON Web Token used for authentication
- **Microservices**: Architecture where the system is divided into small independent services

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2. Overall Description

2.1 Product Perspective

The system follows a microservices architecture where each module operates independently and communicates through REST APIs.

2.2 User Classes

User Type	Description
User	Can browse, purchase, and manage orders
Admin	Manages products, users, and system data

2.3 System Features

- User Management
- Product Management
- Order Management
- Payment System

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3. Functional Requirements

3.1 User Management

- The system shall allow users to register.
- The system shall allow users to log in and log out.
- The system shall allow users to view and update their profile.

3.2 Product Management

- The system shall display all available products.
- The system shall allow searching for products.
- The admin shall be able to add products.
- The admin shall be able to update products.
- The admin shall be able to delete products.

3.3 Order Management

- The system shall allow users to add products to the cart.
- The system shall allow users to place orders.
- The system shall allow users to view order history.

3.4 Payment System

- The system shall process payments (mock or real).
- The system shall store payment details securely.

3.5 Authentication & Authorization

- The system shall use JWT for authentication.
- The system shall support role-based access (Admin and User).

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4. Non-Functional Requirements

4.1 Security

- User passwords must be encrypted.
- Secure authentication must be implemented.

4.2 Performance

- The system should handle multiple users simultaneously.

4.3 Scalability

- The system should support scaling using microservices architecture.

4.4 Availability

- The system should be available 24/7 with minimal downtime.

4.5 Deployment

- The system will be containerized using Docker.

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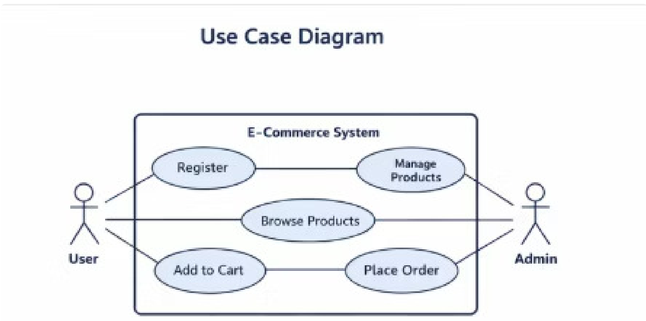
5. System Models (Diagrams)

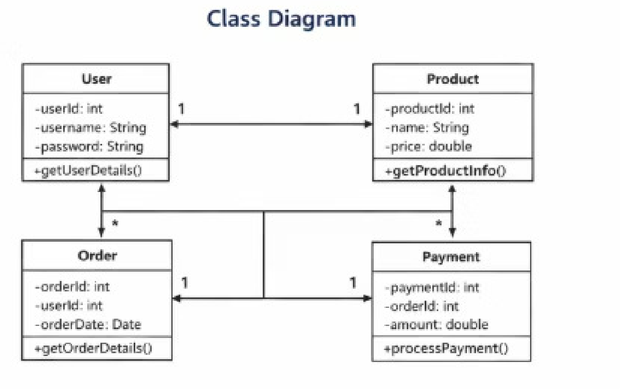
The following diagrams must be included:

5.1 Use Case Diagram



Shows interaction between User and Admin with the system.

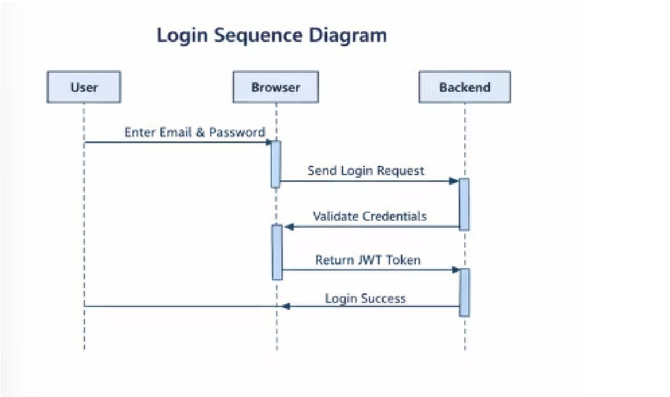
5.2 Class Diagram



Includes:

- User
- Product
- Order
- Payment

5.3 Sequence Diagram



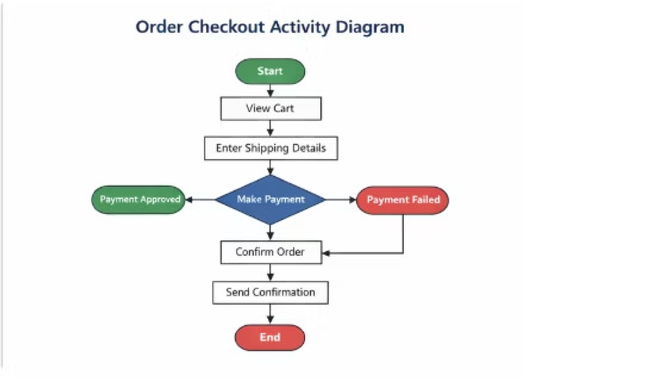
Examples:

- Login Process
- Order Placement

5.4 Activity Diagram

Example:

- Checkout Process



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6. Assumptions and Constraints

6.1 Assumptions

- Users have internet access.
- Payment system may be simulated.

6.2 Constraints

- The system must be built using React and Spring Boot.
- The system must use MySQL database.

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7. Future Enhancements

- Integration with real payment gateways
- Recommendation system
- Mobile application

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8. Conclusion

This document defines the complete requirements for the E-Commerce System and serves as a reference for development and implementation.

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